CS 215: Task 06: Due: 7-Oct; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual assignment.

Teaching-team notes.

- For all questions in this task, explicitly write down the assumptions made.
- For all problems here, do not use electronic calculating aids since these will not be allowed in the exam. You may refer to the tables mentioned in the appendix.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. The time taken by non-hostel resident students to travel to IIT Bombay has mean 42 minutes and standard deviation 12 minutes. Travel times on different days may be treated as independent.

   Suppose the travel times over 64 days are recorded. What is the probability that the average travel time is at least 39 minutes and at most 45 minutes?

2. The amount spent by a typical customer at Cafe92 has mean Rs. 180 and a standard deviation Rs. 90. Amounts spent by different customers may be treated as independent. On a particular evening, 100 customers visit Cafe92. What is the probability in percentage that the total revenue exceeds Rs. 20,000?

3. For each student in a class, independently, the probability of attending an optional review session is 0.62. A total of 400 students are eligible to attend. Using the Central Limit Theorem, approximate the probability that the number of students attending is between 240 and 260, inclusive.

4. The lifetime of an electronic chip has mean 500 hours and standard deviation 20 hours. Lifetimes of different chips may be treated as independent. The distribution of the lifetime of an individual component is not assumed to be normal.

   A student wishes to purchase n independently selected chips. Let $\overline{X}$ denote their average lifetime. Using the Central Limit Theorem, determine the smallest integer n for which

   $$P\left(|\overline{X} - 500| < 5\right) \geq 0.95.$$

   Teaching-team notes.

   $$P(|Z| \leq 1.96) \approx 0.95, \qquad Z \sim \mathcal{N}(0,1).$$

   You are expected to remember this number 1.96 during the exam, much the way you remember $\pi \approx 3.14$. It corresponds to the 95% confidence interval.

5. The weight of a packaged item has mean 100 grams and standard deviation 24 grams. Weights of different items may be treated as independent. A distributor is allowed to return a shipment back to the manufacturer if the average weight of the items in the shipment is more than c since the consumer gets more quantity for the same price.

   A distributor handles a random shipment of 144 items. Let $\overline{X}$ denote the average weight of the items.

   Using the Central Limit Theorem, find the value c such that approximately

   $$P(\overline{X} \leq c) = 0.975.$$

   Teaching-team notes. You may use

   $$P(Z \leq 1.96) = 0.975, \qquad Z \sim \mathcal{N}(0,1).$$

6. Suppose that, independently for each student, the probability that a student uses public transport to commute to campus is 0.40.

   A random sample of 600 students is chosen. Let $\widehat{P}$ denote the proportion of students in the sample who use public transport.

   Using the Central Limit Theorem for proportions, approximate

$$P(0.36 \leq \widehat{P} \leq 0.44).$$

7. For each of the following cases, consider

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For each case, determine whether the Central Limit Theorem as stated in class can be applied to approximate the distribution of $\overline{X}_n$ for large n. If it can, give the approximate distribution. If it cannot, state which assumption is missing. Write your answer starting with a phrase such as "Applies".

(a) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$E[X_i] = 5, \qquad \mathrm{Var}(X_i) = 9.$$

(b) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$E[X_i] = 5, \qquad E[X_i^2] = \infty.$$

(c) The random variables $X_1, \ldots, X_n$ are independent and satisfy

$$E[X_i] = 5, \qquad \mathrm{Var}(X_i) = 9,$$

but they are not assumed to have the same distribution.

(d) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$X_i \sim \mathcal{N}(5, 9).$$

In this case, also state whether the distribution obtained for $\overline{X}_n$ is approximate or exact.

8. Let $X_1, X_2, \ldots$ be i.i.d. random variables satisfying

$$E[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2,$$

and suppose

$$E[(X_i - \mu)^3] = \gamma\sigma^3,$$

where $\gamma \neq 0$.

For

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

define the standardized sample mean

$$Z_n = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma}.$$

(a) Show that

$$E[Z_n] = 0, \qquad \mathrm{Var}(Z_n) = 1.$$

(b) Show that

$$E[Z_n^3] = \frac{\gamma}{\sqrt{n}}.$$

(c) Compute $E[Z_n^3]$ for

$$n = 10, \qquad n = 200, \qquad n = 1000$$

and compare their magnitudes.

(d) Now take five independent batches, each containing 200 iid observations from the same distribution as X_i. Let $\overline{X}_{200}^{(j)}$ be the sample mean of batch j, and define

$$M = \frac{1}{5} \sum_{j=1}^{5} \overline{X}_{200}^{(j)}.$$

Show that M has the same mean, variance, and standardized third moment as a sample mean based on 1000 observations.